

tobler: Spatial interpolation and areal analysis for Python

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Summary

tobler is an open-source Python library for spatial interpolation and areal data transformation, designed to support a wide range of workflows in geographic information science (GIScience), spatial data science, and applied spatial analysis. The package provides a unified, extensible framework for transferring data between incompatible zonal systems—an essential task when working with spatially aggregated data such as census geographies, administrative boundaries, or service areas.

Areal interpolation is a foundational problem in spatial analysis, arising whenever data collected over one set of polygons (source zones) must be reallocated to another (target zones) with different spatial configurations. tobler implements a suite of methods for this task, ranging from simple area-weighted interpolation to more advanced dasymetric and model-based approaches. The package is designed to operate natively on GeoPandas GeoDataFrames, enabling seamless integration into modern Python-based geospatial workflows.

As part of the Python Spatial Analysis Library (PySAL) ecosystem (S. J. Rey et al., 2022; S. J. Rey & Anselin, 2007), tobler adheres to shared design principles emphasizing composability, transparency, and reproducibility. It fills a critical gap in the Python geospatial stack by providing robust, well-tested implementations of areal interpolation methods that are both accessible for applied users and extensible for methodological research.

Statement of need

Spatial data are frequently aggregated to arbitrary or administratively defined units, such as census tracts, ZIP codes, or political boundaries. These units often change over time or differ across datasets, creating substantial challenges for longitudinal analysis, data integration, and cross-scale comparison. Analysts are therefore routinely required to transfer variables from one set of spatial units to another, a process complicated by partial overlaps, mismatched boundaries, and differences in spatial resolution. This problem is especially frequent in the social and behavioral sciences, where data sources like surveys (e.g. the U.S. Census) are published as aggregate statistics within a set of geographically-referenced polygons. While there are methods and Python packages for conducting *continuous* spatial interpolation, commonly applied in the Earth sciences and physical Geography—including examples like PyKrig (Murphy et al., 2025) or pyinterpolate (Moliński, 2022)—such methods are inappropriate for interpolating data between discrete zoning systems.

Traditional GIS software (e.g., ArcGIS, QGIS) provides tools for areal interpolation, but these implementations are often embedded in graphical interfaces, limiting reproducibility and

41 automation. Moreover, they may not expose the full range of methodological options or allow
42 for easy integration into data science pipelines.

43 Prior to `tobler`, the Python ecosystem support for areal interpolation was fragmented or
44 limited. While foundational libraries such as GeoPandas provide data structures and geometric
45 operations, they do not natively implement interpolation methods. As a result, users often
46 relied on *ad hoc* scripts or external tools, leading to inconsistent workflows and potential
47 methodological errors.

48 `tobler` addresses these challenges by providing:

- 49 ▪ A coherent and well-documented API for areal interpolation
- 50 ▪ Native integration with GeoPandas and the broader PySAL ecosystem
- 51 ▪ Support for multiple interpolation paradigms, including area-weighted, dasymetric, and
52 model-based approaches
- 53 ▪ A focus on reproducibility, allowing for version controlled and shared interpolation
54 workflows.

55 These capabilities make `tobler` particularly valuable for researchers and practitioners in fields
56 such as geography, urban planning, public health, environmental science, and regional economics,
57 where spatial data integration is a routine requirement.

58 State of the field

59 `tobler` is a component of the PySAL ecosystem (S. Rey, Stephens, et al., 2026), which
60 provides a comprehensive suite of tools for spatial analysis in Python. This ecosystem consists
61 of four layers of related packages:

- 62 ▪ `libpysal` provides spatial weights, data structures, and foundational utilities (S. Rey,
63 Wolf, et al., 2026)
- 64 ▪ `explore` supports exploratory spatial data analysis, segregation, inequality, geographical
65 dynamics, point patterns, and morphometrics
- 66 ▪ `model` implements spatial econometric models, geographically weighted regression, spatial
67 accessibility, spatial optimization, and geographically weighted machine learning
- 68 ▪ `viz` provides classification schemes for choropleth mapping

69 Within this ecosystem, **`tobler`** occupies a critical niche by providing the data integration and
70 interpolation methods necessary to harmonize datasets before analysis in the **`explore`** or **`model`**
71 phases.

72 `tobler` complements these packages by addressing the specific problem of spatial data
73 transformation between incompatible zonal systems.

74 Compared to desktop GIS platforms, `tobler` offers several advantages:

- 75 ▪ **Reproducibility:** Workflows can be scripted and version-controlled
- 76 ▪ **Transparency:** Methods and assumptions are explicit and inspectable
- 77 ▪ **Extensibility:** Users can modify or extend algorithms for research purposes
- 78 ▪ **Integration:** Interpolation can be embedded within larger data science pipelines, including
79 machine learning and statistical modeling

80 While similar functionality exists in other ecosystems (e.g., R packages such as `areal` (Prener
81 et al., 2019) or `sf`-based workflows (Pebesma, 2018)), `tobler` provides a native solution for
82 Python users, aligning with the growing adoption of Python in geospatial and data science
83 communities.

84 Software design

85 `tobler` is designed with attention to both computational efficiency and usability. Spatial
86 overlay operations, which are central to areal interpolation, can be computationally intensive
87 for large datasets. The package leverages vectorized operations and efficient geometric libraries
88 (via GeoPandas (Bossche et al., 2026) and Shapely (Gillies et al., 2026)) to handle these tasks.

89 The API design emphasizes clarity and consistency, with function signatures that explicitly
90 distinguish between extensive and intensive variables. Intensive variables are things like rates or
91 percentages that must be averaged when aggregating, whereas extensive variables are things
92 like counts that must be summed during aggregation. This design reduces the likelihood
93 of common errors in interpolation workflows and encourages best practices in spatial data
94 handling.

95 Additionally, `tobler` is developed with testing and documentation standards consistent with
96 the Scientific Python ecosystem, ensuring reliability and maintainability.

97 Core Functionality

98 `tobler` organizes its functionality around several key interpolation paradigms, each
99 corresponding to different assumptions about how variables are distributed within source zones.

100 Area-weighted interpolation

101 Area-weighted interpolation is the most basic and widely used method for transferring data
102 between polygon layers. It assumes that variables are uniformly distributed within each source
103 zone and allocates values to target zones in proportion to the area of overlap.

104 `tobler` provides efficient implementations for both **extensive variables** (e.g., population counts)
105 and **intensive variables** (e.g., rates or densities), ensuring appropriate handling of each type
106 (Goodchild & Lam, 1980). The library also supports pycnophylactic adjustments to preserve
107 totals where required (Tobler, 1979).

108 Dasymetric interpolation

109 Dasymetric interpolation refines area-weighted approaches by incorporating ancillary data—such
110 as land use, land cover, or remotely sensed information—to model the internal heterogeneity
111 of source zones. For example, population may be redistributed only to residential areas rather
112 than uniformly across all land (Eicher & Brewer, 2001; Mennis & Hultgren, 2006; Reibel &
113 Agrawal, 2007).

114 `tobler` supports both vector- and raster-based dasymetric workflows, allowing users to integrate
115 a wide range of auxiliary datasets. This is particularly useful in urban and environmental
116 applications where fine-scale heterogeneity is important.

117 Model-based interpolation

118 Beyond deterministic approaches, `tobler` includes model-based methods that use statistical or
119 machine learning techniques to estimate spatial distributions. These approaches incorporate
120 covariates and capture more complex spatial patterns, providing improved accuracy in many
121 contexts (Flowerdew & Aitkin, 1982; Flowerdew & Green, 1992).

122 The design of `tobler` allows these methods to be extended and customized, making the
123 package a useful platform for methodological research in spatial interpolation.

124 Integration with GeoPandas

125 All core functions in `tobler` operate directly on GeoPandas GeoDataFrames, minimizing friction
126 in typical workflows. Users can pass source and target datasets as GeoDataFrames, specify
127 variables of interest, and obtain interpolated results as new GeoDataFrames. This design
128 leverages the broader geospatial Python stack, including Shapely for geometry operations and
129 pandas for tabular data handling.

130 Example workflow

131 A typical area-weighted interpolation in `tobler` is implemented as follows:

```
132 from tobler.area_weighted import area_interpolate  
  
133 result = area_interpolate(  
134     source_df,  
135     target_df,  
136     extensive_variables=["population"],  
137     intensive_variables=["income"]  
138 )
```

132 This operation transfers population counts and income measures from the source geometries
133 to the target geometries, handling each variable type (extensive/intensive) appropriately.

134 When additional information about within-zone heterogeneity is available, dasymetric
135 interpolation can be used to refine estimates. For example, population counts may be
136 redistributed using a land cover raster to exclude uninhabited areas:

```
137 from tobler.dasymetric import masked_area_interpolate  
  
138 result = masked_area_interpolate(  
139     raster="raster_file_name.tif",  
140     source_df,  
141     target_df,  
142     pixel_values = [21,22,23,24],  
143     extensive_variables=["population"]  
144 )
```

137 This approach assumes the user have a raster data of his own that can be read by `rasterio`¹.
138 In this example, `tobler` allows a flexible approach where the user can pass which pixels are to
139 be assumed inhabited through `pixel_values` resulting in a more realistic spatial distribution.
140 Similarly, the user can execute a model-based approach using the `tobler.model.glm` function.

141 [Figure 1](#) illustrates an example comparing interpolated values derived from different spatial
142 configurations, highlighting how results may vary depending on the underlying geometry and
143 interpolation approach.

¹A common example is the ones available at the [National Land Cover Database](#).

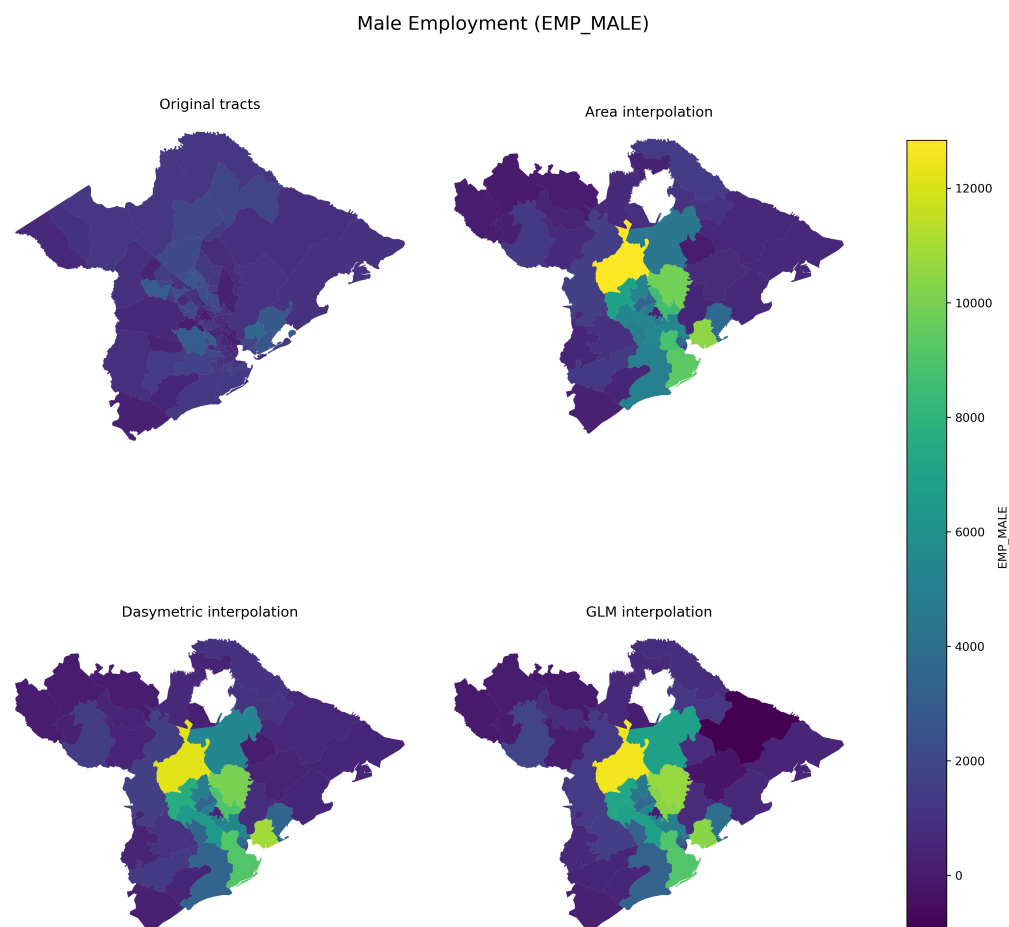


Figure 1: Example of tobler usage for a single extensive variable – employed male population – (no intensive variable) in Charleston, SC, comparing census tracts and ZCTAs.

Research impact statement

The package is actively used by the research community to transfer the data between various types of geographic boundaries. This is not limited to specific applications but covers use cases from continental analysis of emissions and health (La Porta & Zapperi, 2024), analysis of urban form and function (Fleischmann & Arribas-Bel, 2022), redistribution of census data to school districts for assessment of the Clean School Bus Rebate Program (Osia et al., 2025), quantification of radon exposure (Lee et al., 2026), and harmonization of vector and raster data for computer vision tasks (Fleischmann & Arribas-Bel, 2024).

Moreover, the package is relied on in downstream software such as atlasbr for harmonization of Brazilian urban data (Oliveira & Paiva Neto, 2025), and is referred to in the pygridmap package by Eurostat (Grazzini & Gaffuri, 2020) as a reference implementation. The tobler package has made tangible contributions to spatial science, pedagogy, and applications in government and industry. In academia, the package is used as part of a data-processing pipeline for research that examines the spatial-contextual influence on a variety of outcomes, including segregation (Wei et al., 2022), housing policy (S. J. Rey & Knaap, 2022), education policy (Osia et al., 2025; S. J. Rey et al., 2024), and pollution exposure (La Porta & Zapperi, 2024; Lee et al., 2026). It is also used in environmental science (Hu et al., 2023) and regionalization research (Feng et al., 2022).

162 In spatial data science education, `tobler` has become an integral part of many curricula. It is
163 included in popular pedagogical resources, including two textbooks (Knaap, 2026; S. J. Rey
164 et al., 2023), and is taught in graduate and undergraduate courses in universities across the
165 globe, including the University of California (Berkeley, Irvine, Riverside, and Santa Barbara
166 campuses), San Diego State University, Charles University, University of Liverpool, Northern
167 Arizona University, and Temple University, among others.

168 In the public sector, the `tobler` package is used as part of a processing pipeline that powers
169 urban planning and policymaking, including the [DemoLand](#) project from the Alan Turing
170 Institute, developed as a part of the British National Land Data Programme ([Geospatial
171 Commission, 2023](#)). The project piloted predictive modelling based on data interpolated to a
172 unified hexagonal grid and is scaling the approach nation-wide.

173 **AI usage disclosure**

174 No generative AI or LLMs were used for code production for `tobler` or the writing of this
175 paper.

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177 `tobler` is developed as part of the PySAL community, which brings together researchers and
178 developers working on spatial analysis methods and software. The project builds on decades
179 of research in areal interpolation, dasymetric mapping, and spatial data science, and benefits
180 from contributions across the open-source geospatial community.

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